

Table 13: TDI weight sensitivity analysis. Performance (subtask completion %) across weight configurations. Bold indicates selected weights.

w_H	w_E	w_C	w_S	Performance (%)
0.25	0.25	0.25	0.25	71.2
0.30	0.30	0.20	0.20	73.8
0.35	0.25	0.20	0.20	72.4
0.25	0.35	0.20	0.20	73.1
0.30	0.25	0.25	0.20	72.9
0.40	0.30	0.15	0.15	70.8

Table 14: Mode selection threshold sensitivity. Performance (subtask completion %) across threshold configurations.

θ_{explore}	θ_{exploit}	Performance (%)
0.5	0.2	72.1
0.5	0.3	72.8
0.6	0.2	73.2
0.6	0.3	73.8
0.6	0.4	72.4
0.7	0.3	73.0
0.7	0.4	71.6

Table 15: Pruning parameter sensitivity. Metrics: subtask completion (%), branches incorrectly pruned (%), wasted attempts on intractable branches (mean count).

θ_{prune}	k_{min}	Completion	False Prune	Wasted
0.7	2	71.2	8.4	2.1
0.7	3	72.4	5.2	3.4
0.8	3	73.8	2.8	4.1
0.8	4	73.2	1.9	5.8
0.9	3	72.1	1.2	6.9

Table 16: UCB difficulty penalty (λ) sensitivity.

λ	Completion (%)	Backtrack Rate (%)
0.0 (standard UCB)	68.4	12
0.25	71.2	21
0.5	73.8	34
0.75	72.1	42
1.0	69.8	51